

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EC)

May 2017

EC 308.01 Antenna & Wave Propagation

Date: 30.05.2017, Tuesday Time: 10.00 a.m. To 1.00 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION -I

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|-----------|--------|--|----|
| Q-1 | a. | The measured HPBW of an antenna in the two orthogonal planes are 30° and 20° . Antenna efficiency 100%. The approximate gain of the antenna is : _____ | 1 |
| | b. | If antenna directivity and antenna gain are equal, then antenna efficiency is _____%. | 1 |
| | c. | The radiation resistance of antenna is 80 ohm and loss resistance is 10 ohm. What is the directivity, if power gain is 20? | 1 |
| | d. | Calculate the front to back ratio of an antenna in dB which radiates 3 kW in its most optimum direction and 500 watts in opposite direction. | 1 |
| | e. | The radiation resistance of half wave dipole is _____. | 1 |
| | f. | Explain
Skip Distance, Radiation pattern | 2 |
| Q-2 | a. | Design Rhombic antenna (leg length, tilt angle, height above ground) at frequency 10 GHz and β is 25° . | 2 |
| Q-2 | b.(1) | Sketch the helical geometry with its associated dimension showing relationship between circumferences, spacing, turn length and pitch angle of helix. Also discuss practical design consideration. | 6 |
| | b. (2) | Draw different types of horn antenna and explain them in detail. | 6 |
| OR | | | |
| Q-2 | b. | Starting from retarded currents derive an expression for E and H components for a short dipole antenna if the spherical system is defined in r , θ and Φ . | 12 |
| Q-3 | a. | Determine the distance from short dipole operating at 1MHz at which radiation field is equal to the induction field | 2 |
| Q-3 | b. | Derive the E field equation of array two point sources with equal amplitude and opposite phase also draw radiation pattern. | 6 |
| | c. | Compare V antenna and rhombic antenna | 6 |
| OR | | | |
| Q-3 | b. | Write the statement of pattern multiplication theorem. Explain it in detail by using any example. | 6 |
| | c. | Explain in detail different types of propagations. | 6 |

SECTION II

Q-4	a.	If the impedance of dipole antenna is 73 ohm then what is the impedance of its complementary antenna?	2
	b.	If dipole is horizontally polarized than its complementary antenna is _____polarized.	1
	c.	As the number of turns of an axial mode helical antenna decreases, its beam-width_____.	1
	d.	A microstrip antenna with overall dimensions of $L = 0.906$ cm (0.357 inches) and $W = 1.186$ cm (0.467 inches), substrate with height $h = 0.1588$ cm (0.0625 inches) and dielectric constant of $\epsilon_r = 2.2$, is operating at 10 GHz and $R_{in} = 228$ ohm. Find the position of the inset feed point where the input impedance is 50 ohms.	2
	e.	What is multipath fading?	1
Q-5	a.	Explain Rumsay's principle for frequency independent antenna.	2
	b.	Find radiation resistance of single loop antenna if $c/\lambda = 0.1$.	2
Q-5	c.(1)	A pyramidal horn with E plane aperture $a_E = 10\lambda$ is fed by rectangular waveguide with TE ₁₀ mode. Let $\delta = 0.2\lambda$ in E plane and 0.375λ in H plane. Find following parameters. Length of horn, H plane aperture flare angle in E plane and H plane. HPBW for E plane and H plane.	4
	c.(2)	Draw and explain structure of Log periodic antenna. Derive its mathematical equations for designing purpose.	6
OR			
Q-5	c.(1)	Explain the experimental setup for the measurement of Radiation pattern of antenna.	5
	c.(2)	Design 6 element yagi uda arrays to provide gain of 12dBi if operating frequency is 500MHz.	5
Q-6	a.	Explain different Feed Mechanism used in parabolic Reflector antenna systems.	5
	b.	Design rectangular patch antenna at 2.4GHz dielectric constant 4.4 and thickness of dielectric material is 1.6mm.	5
	c.	Derive equation of received power of antenna (FRISS formula).	4
OR			
Q-6	a.	Draw and explain different types of reflectors	5
	b.	Explain Babinet's principle and discuss it for complementary antenna.	5
	c.	Explain different feed mechanism used in patch antenna	4